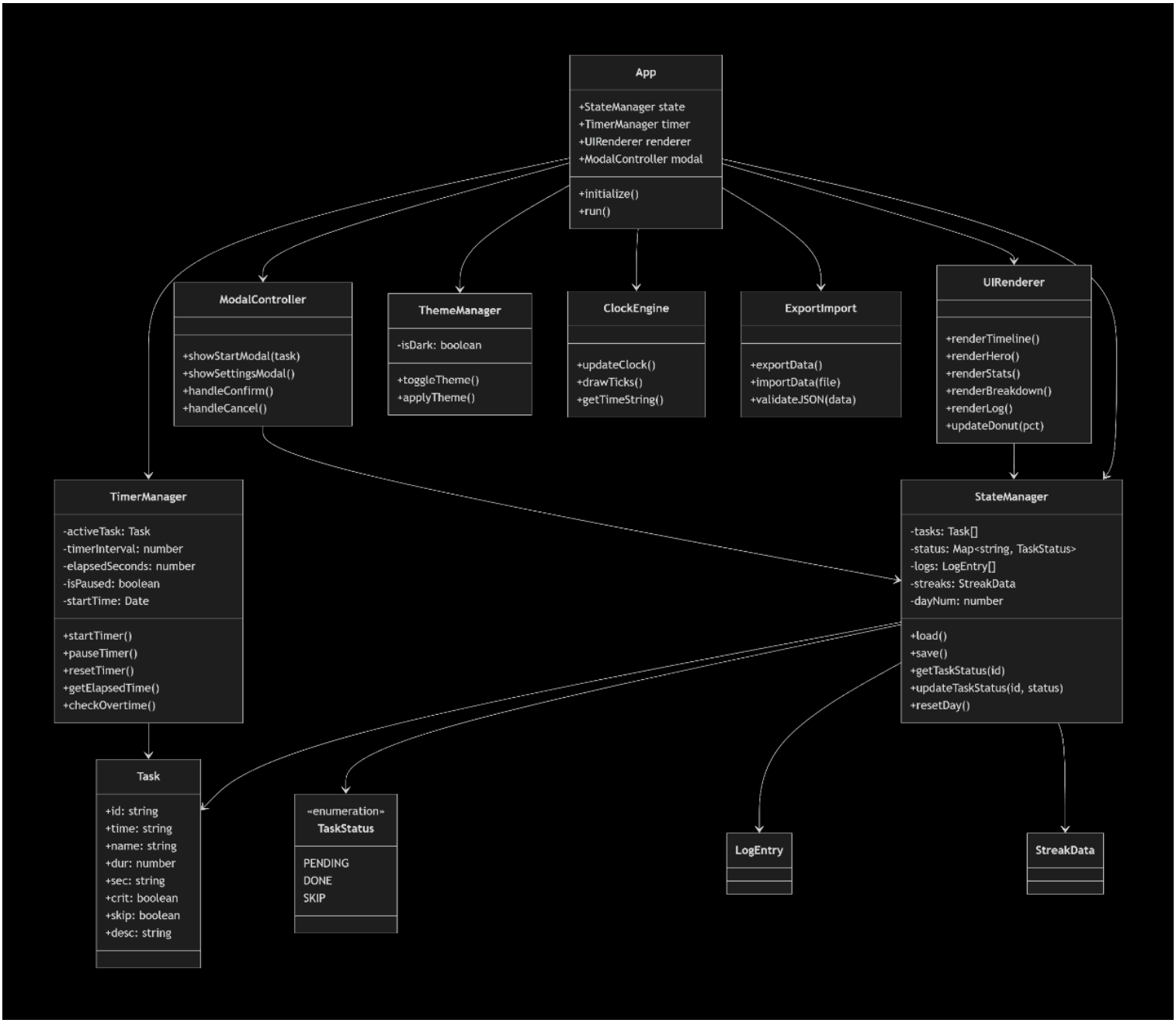
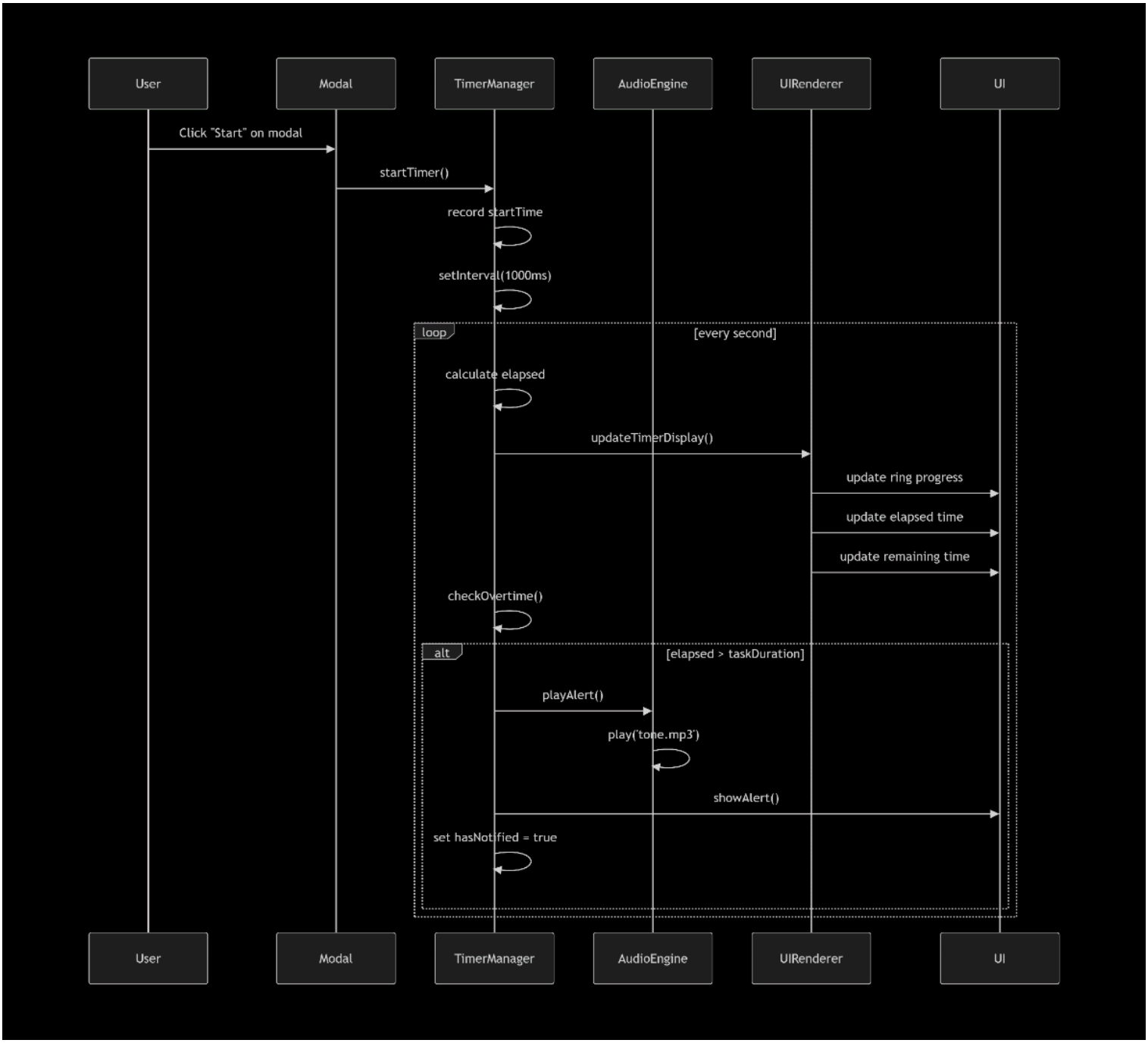
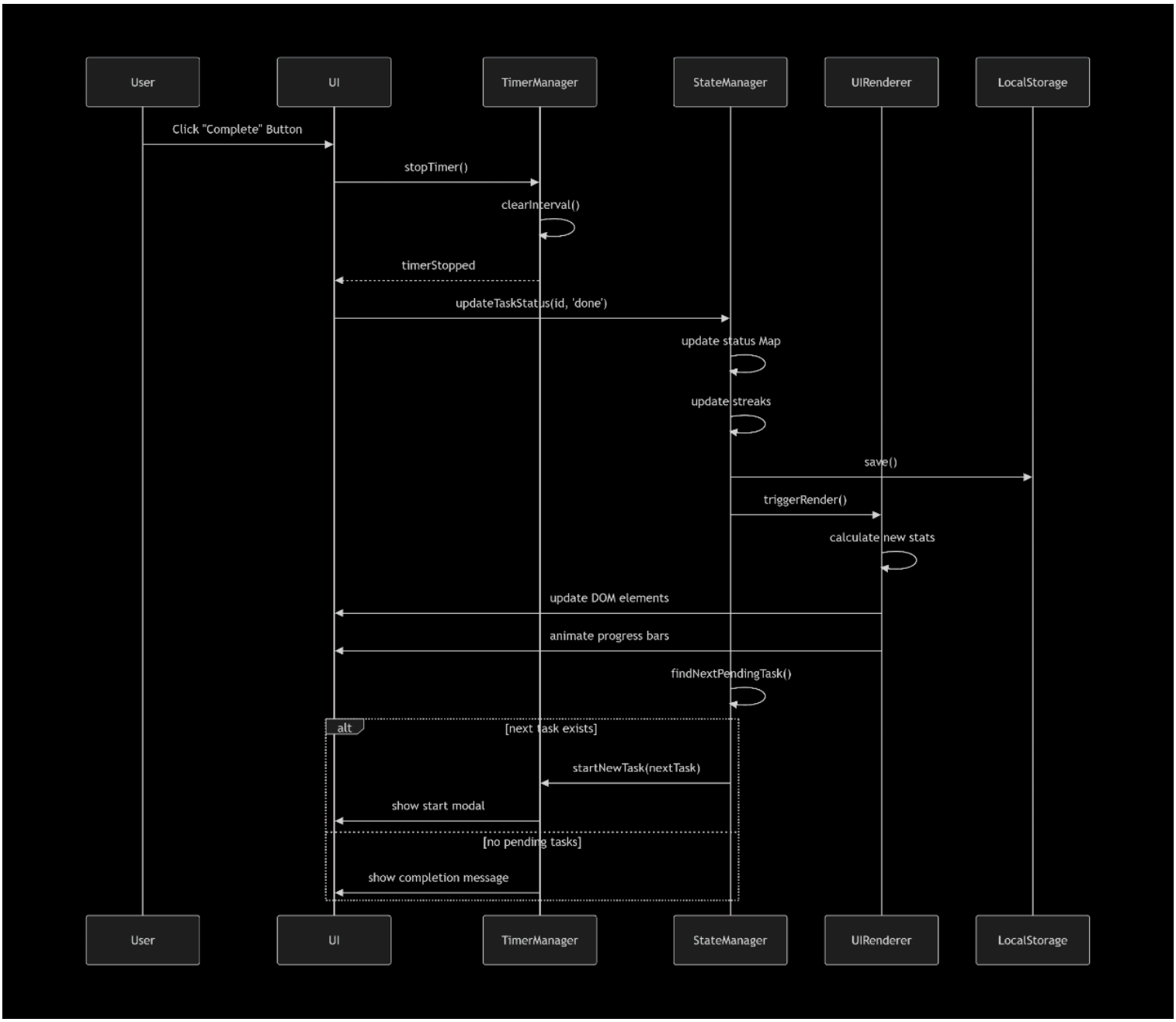
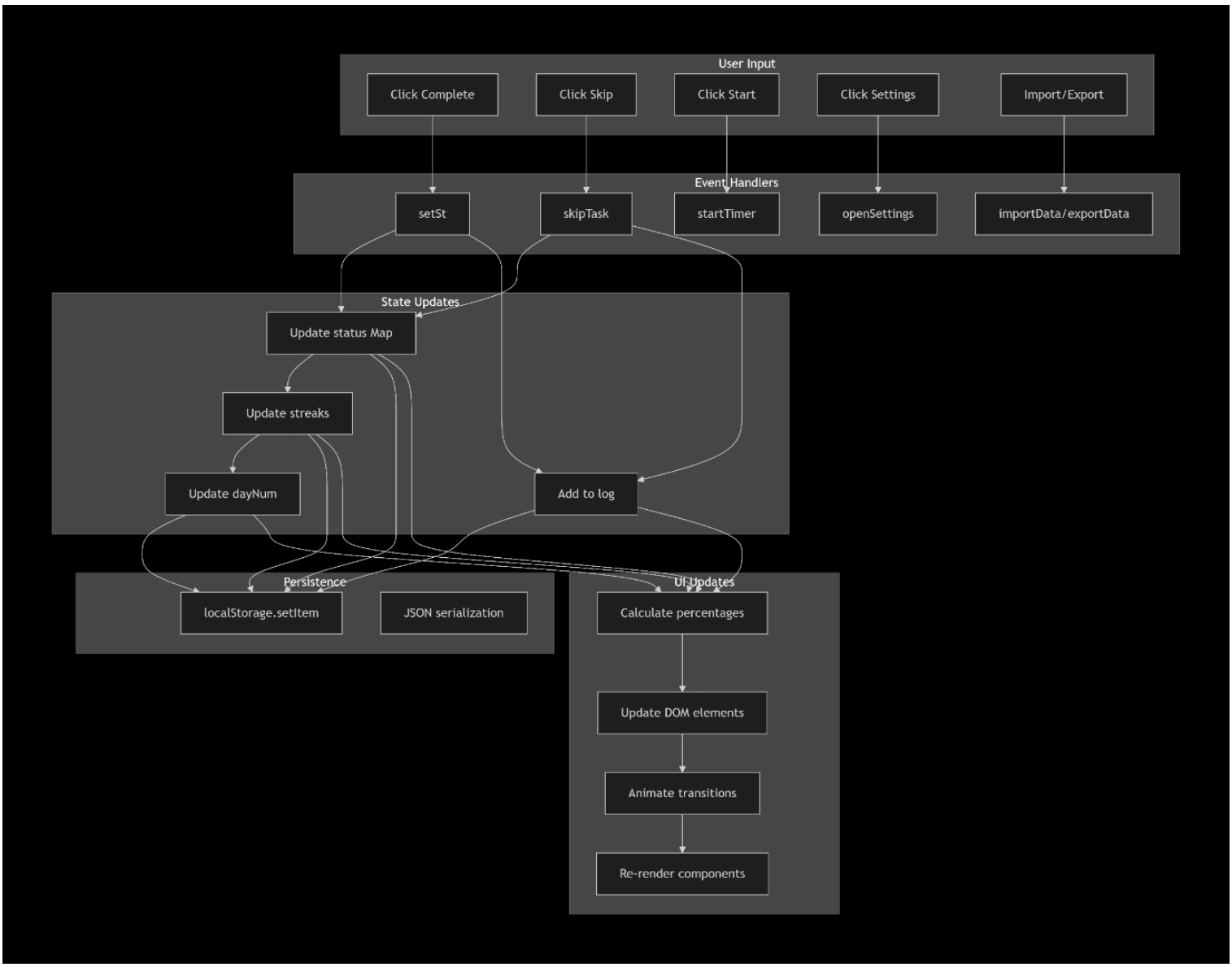
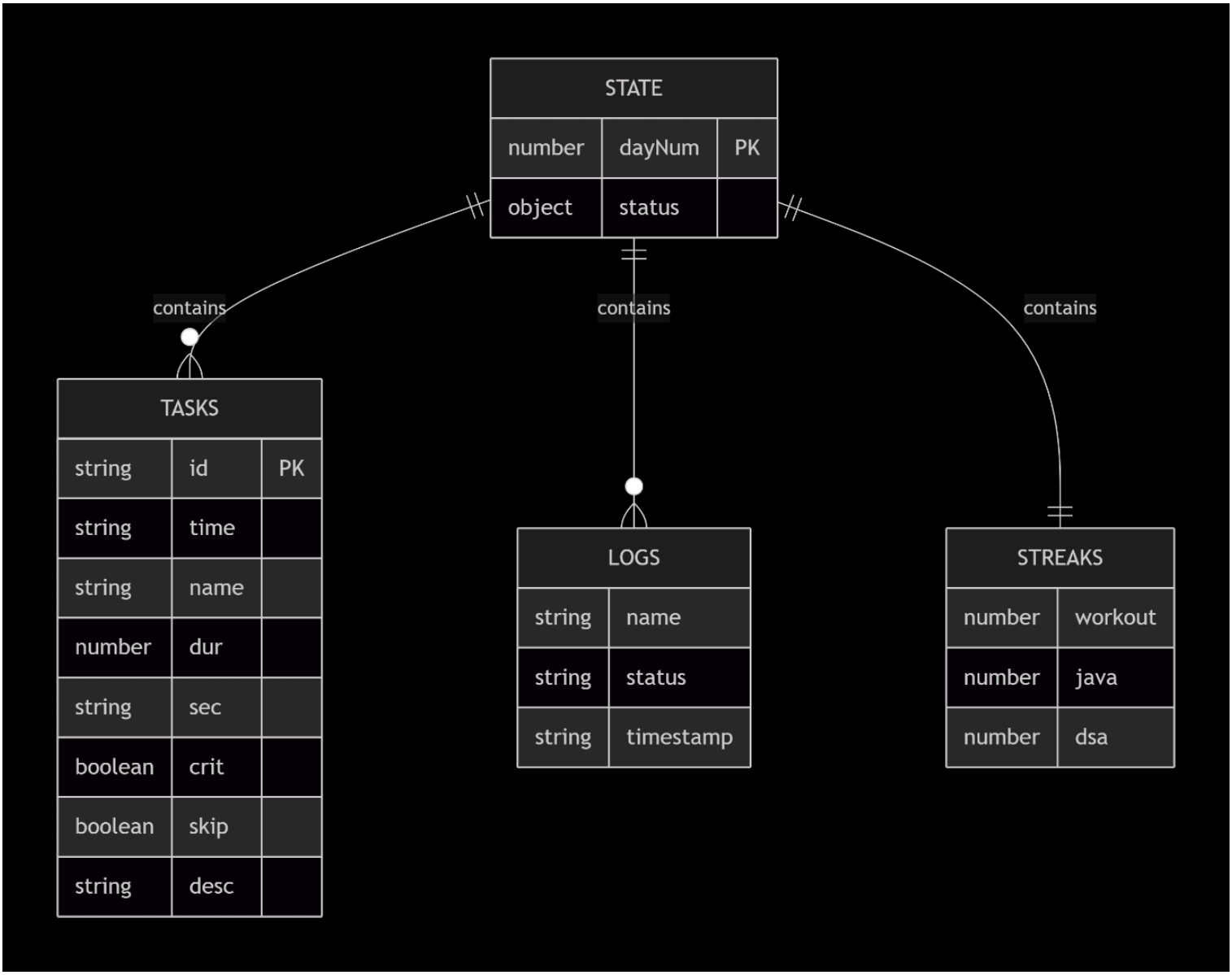

App
+StateManager state
+TimerManager timer
+UIRenderer renderer
+ModalController modal
+initialize()
+run()
ModalController
+showStartModal(task)
+showSettingsModal()
+handleConfirm()
+handleCancel()
ThemeManager
-isDark: boolean
+toggleTheme()
+applyTheme()
ClockEngine
+updateClock()
+drawTicks()
+getTimeString()
ExportImport
+exportData()
+importData(file)
+validateJSON(data)
UIRenderer
+renderTimeline()
+renderHero()
+renderStats()
+renderBreakdown()
+renderLog()
+updateDonut(pct)
TimerManager
-activeTask: Task
-timerInterval: number
-elapsedSeconds: number
-isPaused: boolean
-startTime: Date
+startTimer()
+pauseTimer()
+resetTimer()
+getElapsedTime()
+checkOvertime()
StateManager
-tasks: Task[]
-status: Map<string, TaskStatus>
-logs: LogEntry[]
-streaks: StreakData
-dayNum: number
+load()
+save()
+getTaskStatus(id)
+updateTaskStatus(id, status)
+resetDay()
Task
+id: string
+time: string
+name: string
+dur: number
+sec: string
+crit: boolean
+skip: boolean
+desc: string
«enumeration»
TaskStatus
PENDING
DONE
SKIP
LogEntry
StreakData


User
Modal
TimerManager
AudioEngine
UIRenderer
UI
Click "Start" on modal
startTimer()
record startTime
setInterval(1000ms)
loop
[every second]
calculate elapsed
updateTimerDisplay()
update ring progress
update elapsed time
update remaining time
checkOvertime()
alt
[elapsed > taskDuration]
playAlert()
play("tone.mp3")
showAlert()
set hasNotified = true
User
Modal
TimerManager
AudioEngine
UIRenderer
UI


User
UI
TimerManager
StateManager
UIRenderer
LocalStorage
Click "Complete" Button
stopTimer()
clearInterval()
timerStopped
updateTaskStatus(id, 'done')
update status Map
update streaks
save()
triggerRender()
calculate new stats
update DOM elements
animate progress bars
findNextPendingTask()
alt
[next task exists]
startNewTask(nextTask)
show start modal
[no pending tasks]
show completion message
User
UI
TimerManager
StateManager
UIRenderer
LocalStorage


User Input
Click Complete
Click Skip
Click Start
Click Settings
Import/Export
Event Handlers
setSt
skipTask
startTimer
openSettings
importData/exportData
State Updates
Update status Map
Update streaks
Update dayNum
Add to log
Persistence
localStorage.setItem
JSON serialization
UI Updates
Calculate percentages
Update DOM elements
Animate transitions
Re-render components


STATE
number
dayNum
PK
object
status
contains
contains
contains
TASKS
string
id
PK
string
time
string
name
number
dur
string
sec
boolean
crit
boolean
skip
string
desc
LOGS
string
name
string
status
string
timestamp
STREAKS
number
workout
number
java
number
dsa